

Marks Average Demonstration

Title: Marks Average Calculation using Scanner

Objective/Aim:

To write a Java program to accept marks of five subjects from the user, calculate the total marks, and compute the average marks using the Scanner class[cite: 12].

System Requirements :

Hardware

- Standard PC/Laptop[cite: 1]
- Intel Core i3/i5 processor or equivalent[cite: 1, 12]
- Minimum 4 GB / 8 GB RAM[cite: 1, 12]
- Minimum 500 MB free storage[cite: 1]
- Keyboard and Monitor[cite: 1]

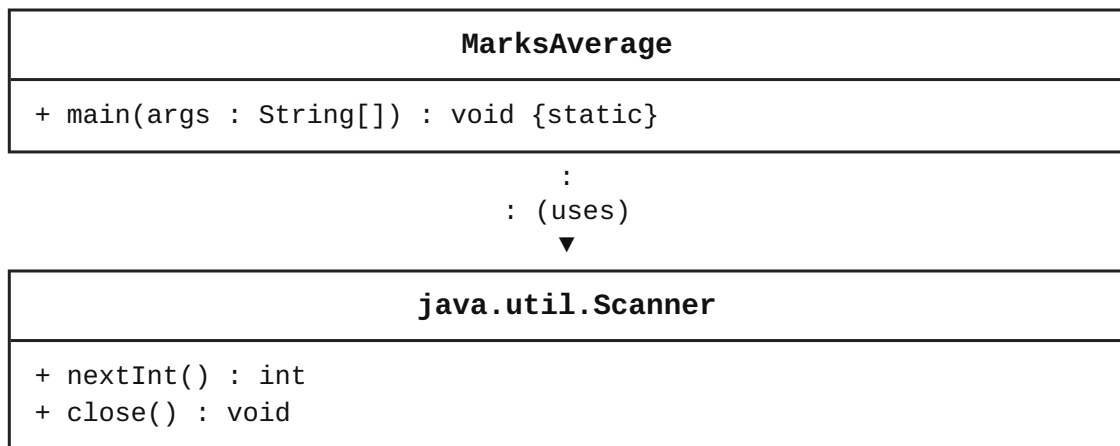
Software

- Operating System: Windows/Linux/macOS[cite: 1, 12]
- Java Development Kit (JDK 17 or JDK 21)[cite: 1, 12]
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code[cite: 1, 12]
- Command Prompt/Terminal[cite: 1]

Algorithm / Logic Description:

1. Import the Scanner class from the `java.util` package[cite: 12].
2. Create a class named `MarksAverage` containing the `main ( )` method[cite: 12].
3. Create a Scanner object named `sc` connected to standard input (`System.in`)[cite: 12].
4. Prompt the user and read the marks of five subjects (`m1`, `m2`, `m3`, `m4`, `m5`) using `nextInt ( )`[cite: 12].
5. Calculate the total marks: `total = m1 + m2 + m3 + m4 + m5`[cite: 12, 13].
6. Calculate the average marks: `average = total / 5.0`[cite: 12, 13].
7. Display the total marks and average marks on the console[cite: 12, 13].
8. Close the Scanner object[cite: 12, 13].
9. Stop the program[cite: 1].

UML class Diagram



Relationship: The `MarksAverage` class depends on and uses the `java.util.Scanner` class to take inputs from the user[cite: 12].

Source Code:

```
import java.util.Scanner;

public class MarksAverage {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter marks of Subject 1: ");
        int m1 = sc.nextInt();

        System.out.print("Enter marks of Subject 2: ");
        int m2 = sc.nextInt();

        System.out.print("Enter marks of Subject 3: ");
        int m3 = sc.nextInt();

        System.out.print("Enter marks of Subject 4: ");
        int m4 = sc.nextInt();

        System.out.print("Enter marks of Subject 5: ");
        int m5 = sc.nextInt();

        int total = m1 + m2 + m3 + m4 + m5;
        double average = total / 5.0;

        System.out.println("Total Marks = " + total);
        System.out.println("Average Marks = " + average);

        sc.close();
    }
}
```

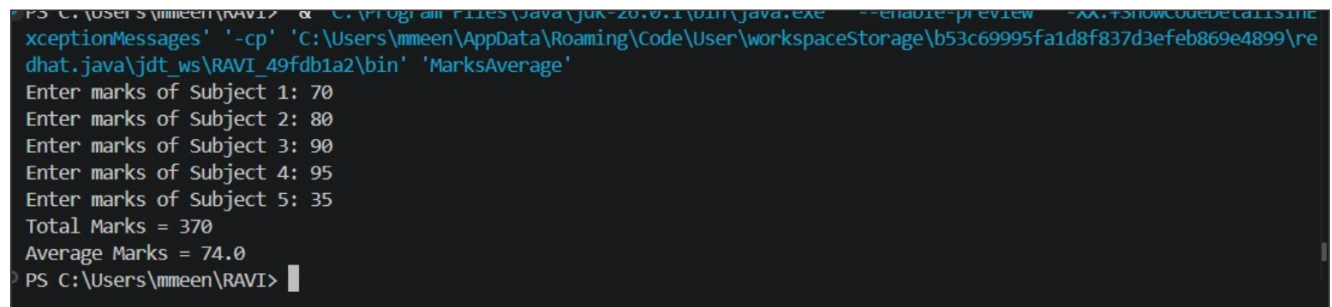
Compilation & Execution Commands:

Save file as: MarksAverage.java[cite: 13]

Compilation: javac MarksAverage.java[cite: 13]

Execution: java MarksAverage[cite: 13]

Output Screenshot:

A screenshot of a Windows command prompt window. The title bar shows the path 'C:\Users\mmeen\RAVI' and the application 'C:\Program Files\Java\jdk-20.0.1\bin\java.exe'. The command prompt shows the execution of the 'MarksAverage' program. The user enters marks for five subjects: 70, 80, 90, 95, and 35. The program outputs the total marks as 370 and the average marks as 74.0. The prompt ends with 'PS C:\Users\mmeen\RAVI>'.

```
PS C:\Users\mmeen\RAVI> & C:\Program Files\Java\jdk-20.0.1\bin\java.exe --enable-preview -XX:+ShowCodeDetailsInExceptionMessages' -cp 'C:\Users\mmeen\AppData\Roaming\Code\User\workspaceStorage\b53c69995fa1d8f837d3efeb869e4899\re
dhat.java\jdt_ws\RAVI_49fdb1a2\bin' 'MarksAverage'
Enter marks of Subject 1: 70
Enter marks of Subject 2: 80
Enter marks of Subject 3: 90
Enter marks of Subject 4: 95
Enter marks of Subject 5: 35
Total Marks = 370
Average Marks = 74.0
PS C:\Users\mmeen\RAVI>
```

Result:

The Marks Average program was successfully implemented in Java. The program demonstrates reading user inputs via the Scanner class, performing arithmetic calculations to determine the total marks, and computing floating-point average marks[cite: 12, 13].